Enhanced Underwater Image Restoration via Sea-Thru Pre-processing and Super-Resolution A Novelty Extension of FUnIE-GAN CSL7360 – Computer Vision Project Report

Group Name: GaMa.CV

Charitha (Roll No: B23EE1039)
Mallam Vishnu Priya (Roll No: B23EE1040)

Abstract

Underwater image enhancement remains a challenging computer vision problem due to the complex optical distortions caused by wavelength-dependent light attenuation, scattering, and refraction. This project builds upon FUnIE-GAN [1], a conditional generative adversarial network designed for fast real-time underwater image enhancement. We propose two novelty extensions:

(1) **Novelty 1** — a physics-guided pre-processing stage using a simplified Sea-Thru [2] attenuation correction combined with adaptive white balancing, applied *before* FUnIE-GAN inference; and

(2) **Novelty 2** — a complete three-stage pipeline that further applies Real-ESRGAN [3] super-resolution to the GAN-enhanced output for higher perceptual fidelity.

Experiments are conducted on two datasets: the paired EUVP dataset [1] (seen, known distribution) and the UIEB benchmark [4] (unseen, unknown distribution). Quantitative no-reference metrics — UIQM, colorfulness, contrast, sharpness, and entropy — along with qualitative five-way comparison strips are used to evaluate all pipeline stages. Results show that the physics-guided preprocessing reduces residual colour casts and that the downstream super-resolution stage produces noticeably sharper textures. On the unseen UIEB dataset, Novelty 2 achieves the highest sharpness score (1826.84) and Novelty 1 achieves the highest UIQM (12.93); on the seen EUVP dataset, Novelty 2 achieves both the highest UIQM (21.65) and sharpness (1919.05). All relevant links to datasets, code, and the original paper are listed in Section 7.

1 Introduction

Underwater robotics and remote sensing increasingly rely on visual perception for tasks such as coral reef monitoring, shipwreck inspection, seabed mapping, and human-robot collaboration in aquatic environments [1]. Despite advances in camera technology, images captured beneath the water surface suffer from severe quality degradation. Light absorption causes red wavelengths to vanish at shallow depths while blue and green channels dominate, producing characteristic blue-green hue casts. Scattering diffuses light and reduces contrast, and turbidity introduces haze-like veiling that obscures scene details. These artefacts degrade the performance of downstream vision pipelines such as object detection, segmentation, and pose estimation.

Two broad families of approaches have been proposed to address these problems. *Physics-based* methods model the formation of underwater images and explicitly estimate medium transmission and ambient illumination to recover scene radiance. While theoretically grounded, these methods typically require per-image tuning and access to scene depth or water optical property measurements that are unavailable in most practical deployments. *Learning-based* methods, by contrast, learn a mapping from degraded to enhanced images from paired or unpaired training data, achieving faster inference and stronger generalisation across diverse underwater conditions.

FUnIE-GAN [1] is a landmark learning-based model that achieves compelling enhancement quality at real-time frame rates. It employs a U-Net-style conditional GAN trained with a multi-modal perceptual loss covering global similarity, image content, and local texture. Despite its strengths, FUnIE-GAN operates solely in the learned domain and does not explicitly account for the physics of underwater light attenuation.

This project investigates whether injecting a lightweight physics-guided correction stage *before* FUnIE-GAN, and a super-resolution stage *afterward*, yields measurable perceptual gains. We evaluate on two datasets to assess both in-distribution performance and generalisation: the EUVP dataset [1] (the training set of FUnIE-GAN, referred to hereafter as the *seen* dataset) and the UIEB benchmark [4] (an entirely independent collection, referred to as the *unseen* dataset).

Summary of contributions:

- A simplified Sea-Thru-inspired preprocessing module combining dark-channel-based depth estimation, adaptive white balancing, and per-channel attenuation correction.
- A full three-stage pipeline integrating the above preprocessing, FUnIE-GAN enhancement, and Real-ESRGAN super-resolution.
- Quantitative evaluation using five no-reference image quality metrics (UIQM, entropy, colorfulness, contrast, sharpness) across four pipeline configurations on both seen and unseen benchmark datasets.
- Qualitative side-by-side comparisons across five pipeline configurations on images from both datasets.
- A reproducible codebase that can serve as a foundation for future hybrid physics-learning underwater enhancement systems.

2 Literature Review

2.1 Physics-Based Underwater Image Enhancement

Classical physics-based approaches to underwater image restoration adapt the atmospheric haze model to the underwater setting. He *et al.*'s dark channel prior [5] is an influential technique for haze removal that exploits the statistical observation that most natural image patches contain at least one colour channel with very low intensity. Adaptations of this prior to the underwater domain achieved moderate colour correction and dehazing.

Akkaynak and Treibitz [2] proposed a revised underwater image formation model that more accurately accounts for the range-dependent and wavelength-dependent nature of underwater light attenuation — phenomena not captured by the atmospheric model. Their Sea-Thru algorithm, built on this revised model, recovers accurate colours given known or estimated scene depth. While effective, the method requires stereo or structured-light depth measurements.

Berman *et al.* [4] introduced a haze-line approach to single image underwater colour restoration, leveraging the clustering of pixels in RGB space along haze lines to estimate the transmission map without explicit depth input. Their accompanying UIEB benchmark provides 890 real underwater images with corresponding reference images and serves as our unseen evaluation set (see Section 7 for the dataset link).

Other physics-guided methods use bilateral and trilateral filtering to reduce noise while preserving contrast, and multi-band fusion frameworks that blend multiple exposures or tone-mapped versions of the same image to produce better contrast and detail rendition.

2.2 GAN-Based Underwater Image Enhancement

Generative adversarial networks have driven substantial progress in underwater image enhancement. Fabbri *et al.* [7] (UGAN-P) proposed an underwater GAN with a gradient penalty term to stabilise training and achieve colour-consistent enhancement. Li *et al.* [8] proposed WaterGAN for unsupervised monocular colour correction. Yu *et al.* [9] introduced Underwater-GAN based on conditional GANs for colour restoration.

FUnIE-GAN [1] consolidated these ideas into a highly efficient model specifically designed for real-time deployment on embedded robotic platforms. Its multi-modal loss function — combining adversarial loss, L1 global similarity, and VGG-19 content loss — yielded state-of-the-art PSNR (21.92 dB) and SSIM (0.8876) on the paired EUVP dataset while running at 25.4 FPS on an NVIDIA Jetson TX2. FUnIE-GAN-UP, its unpaired training variant using CycleGAN-style cycle-consistency, further extended applicability to settings where ground-truth pairs are unavailable. The EUVP dataset contains over 12K paired and 8K unpaired underwater images captured under diverse visibility conditions using seven cameras including GoPro, AUV uEye cameras, and Trident ROV cameras.

2.3 Super-Resolution for Perceptual Quality

Image super-resolution aims to recover fine detail beyond the resolution of the captured image. Wang *et al.* introduced Real-ESRGAN [3], which trains a Residual-in-Residual Dense Block (RRDB) network on a rich set of synthetic degradations to enable robust blind super-resolution of real-world images. Unlike classical bicubic upsampling, Real-ESRGAN recovers sharp textures and realistic high-frequency detail, making it a natural post-processing step for underwater imagery where scattering and absorption have blurred fine structures.

2.4 Hybrid Pipeline Approaches

The combination of physics-based preprocessing and learned enhancement is relatively unexplored for underwater imagery. Cho *et al.* [10] proposed model-assisted multi-band fusion, and Lu *et al.* [11] combined guided filtering with colour correction. Our work is positioned in this hybrid space: we use a lightweight physics-inspired correction to normalise input statistics before a learned GAN, and optionally apply super-resolution to maximise perceptual quality.

2.5 Evaluation Metrics for Underwater Imagery

Quantitative evaluation of underwater image enhancement is complicated by the absence of ground-truth references in many real-world scenarios. The Underwater Image Quality Measure (UIQM) [6] is a purpose-designed composite metric that combines colorfulness (UICM), sharpness (UISM), and contrast (UIConM) into a single scalar. Alongside UIQM, no-reference metrics such as Laplacian variance (sharpness), image entropy, and colorfulness provide complementary views of enhancement quality. When paired reference images are available, full-reference metrics PSNR and SSIM remain the dominant choices, though they can disagree with perceptual quality judgements when textures are hallucinated by generative models.

3 Problem Statement

Let $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ denote a raw underwater image captured under degraded visibility conditions. The image $\mathbf{I}$ is a corrupted observation of the scene's true radiance $\mathbf{J}$ due to wavelength-dependent attenuation, backscattering, and wavelength absorption effects of the water column. Formally, a simplified underwater image formation model can be written as:

$$\mathbf{I}^c = \mathbf{J}^c \cdot t^c(d) + \mathbf{B}^c \cdot \big(1 - t^c(d)\big), \tag{1}$$

where $c \in \{R, G, B\}$ indexes the colour channel, $t^c(d) = e^{-\beta^c d}$ is the channel-dependent transmission factor at scene depth d, β^c is the attenuation coefficient for channel c, and $\mathbf{B}^c$ is the background light (veiling illumination) for channel c.

The goal is to produce an enhanced image $\hat{\mathbf{J}}$ that approximates the true scene radiance $\mathbf{J}$ in terms of perceptual quality — correct colour, improved contrast, and sharp texture — without requiring explicit knowledge of d or β^c.

Our approach decomposes this into three sequential sub-problems:

1. **Physics-guided pre-correction.** Estimate and partially correct the attenuation and colour cast using lightweight image statistics, producing an intermediate image $\mathbf{I}'$ that approximates the output of a Sea-Thru-style correction.
2. **Learned perceptual enhancement.** Apply FUnIE-GAN to map $\mathbf{I}'$ to a perceptually enhanced image $\hat{\mathbf{J}}_{\text{GAN}}$.
3. **Super-resolution (optional).** Apply Real-ESRGAN to $\hat{\mathbf{J}}_{\text{GAN}}$ to recover high-frequency detail, producing the final output $\hat{\mathbf{J}}_{\text{SR}}$.

The key challenge is ensuring that each stage improves rather than degrades the final output. In particular, the preprocessing must be conservative enough to avoid introducing new artefacts while still reducing the input distribution shift that the GAN must handle. Additionally, because FUnIE-GAN was trained specifically on EUVP data, the problem of *domain generalisation* to unseen datasets (UIEB) is a central concern: the preprocessing stage may help by reducing dataset-specific colour biases before GAN inference.

4 Proposed Method

4.1 Stage 1: Sea-Thru-Inspired Preprocessing

Our preprocessing module implements a three-step pipeline inspired by the Sea-Thru algorithm but adapted for scenarios where depth is unavailable.

4.1.1 Adaptive White Balance

We compute the per-channel mean intensities μ_B, μ_G, μ_R and the grey-world estimate $\mu_{\text{gray}} = (\mu_B + \mu_G + \mu_R)/3$. Each channel is scaled by a correction factor:

$$s^c = \text{clip}\left(\frac{\mu_{\text{gray}}}{\mu^c + \epsilon},\ 0.85,\ 1.30\right). \tag{2}$$

The clip bounds $[0.85,\ 1.30]$ are intentionally conservative to prevent over-correction of images that are already nearly colour-balanced. This gentle white balance reduces the dominant blue-green cast common in shallow underwater scenes.

4.1.2 Depth Estimation via Dark Channel Prior

A proxy depth map is estimated using a simplified dark channel prior. For each pixel location (x, y), the dark channel value is:

$$d_{\text{dark}}(x, y) = \min_{c} \min_{(x',y') \in \Omega(x,y)} \mathbf{I}^c(x', y'), \tag{3}$$

where $\Omega(x, y)$ denotes a 15×15 local patch. The depth proxy is defined as $d(x, y) = 1 - d_{\text{dark}}(x, y)$, then Gaussian-blurred and normalised to $[0, 1]$. Although this is not a metric depth, it provides a spatially-varying signal that correlates with the degree of attenuation: darker regions (deeper or more attenuated) receive a higher correction.

4.1.3 Per-Channel Attenuation Correction

Given the depth proxy d, we apply Beer–Lambert-inspired attenuation correction independently per channel:

$$\hat{J}^c(x,y) = \frac{I^c(x,y)}{t^c(x,y)}, \qquad t^c(x,y) = \text{clip}\Big(e^{-\beta^c \cdot d(x,y)},\ 0.75,\ 1.0\Big). \tag{4}$$

We use $\beta_R = 0.15$, $\beta_G = 0.08$, $\beta_B = 0.03$, reflecting the physical reality that red light is most strongly absorbed in water and blue is absorbed least. The transmission floor of 0.75 ensures that no channel is boosted by more than $1/0.75 \approx 1.33\times$, preventing noise amplification in very dark regions. The full implementation is given in Listing 1.

Listing 1: Sea-Thru preprocessing module (`sea_thru.py`).

```
import cv2, numpy as np

def estimate_depth(img, patch_size=15):
    img_f = img.astype(np.float32) / 255.0
    dark = np.min(img_f, axis=2)
    kernel = cv2.getStructuringElement(
        cv2.MORPH_RECT, (patch_size, patch_size))
    dark_ch = cv2.erode(dark, kernel)
    depth = 1.0 - dark_ch
    depth = cv2.GaussianBlur(depth, (21, 21), 0)
    depth = (depth - depth.min()) / (depth.max() - depth.min() + 1e
        -8)
    return depth

def white_balance(img):
    img_f = img.astype(np.float32)
    mean_b = np.mean(img_f[:, :, 0])
    mean_g = np.mean(img_f[:, :, 1])
    mean_r = np.mean(img_f[:, :, 2])
    mean_gray = (mean_b + mean_g + mean_r) / 3.0
    def scale(mean_c):
        return np.clip(mean_gray / (mean_c + 1e-8), 0.85, 1.3)
    img_f[:, :, 0] = np.clip(img_f[:, :, 0] * scale(mean_b), 0, 255)
    img_f[:, :, 1] = np.clip(img_f[:, :, 1] * scale(mean_g), 0, 255)
    img_f[:, :, 2] = np.clip(img_f[:, :, 2] * scale(mean_r), 0, 255)
    return img_f.astype(np.uint8)

def correct_attenuation(img, depth):
    img_f = img.astype(np.float32) / 255.0
    betas = [0.15, 0.08, 0.03]  # R, G, B channels
    J = np.zeros_like(img_f)
    for c, beta in enumerate(betas):
        t = np.clip(np.exp(-beta * depth), 0.75, 1.0)
        J[:, :, c] = img_f[:, :, c] / t
    return (np.clip(J, 0, 1) * 255).astype(np.uint8)

def sea_thru_correct(img_bgr):
    balanced  = white_balance(img_bgr)
    depth     = estimate_depth(balanced)
    corrected = correct_attenuation(balanced, depth)
    return corrected, depth
```

4.2 Stage 2: FUnIE-GAN Enhancement

After Sea-Thru preprocessing, the corrected image $\mathbf{I}'$ is resized to 256×256 and normalised to $[-1, 1]$ before being passed through the pretrained FUnIE-GAN generator. The generator follows a U-Net encoder-decoder architecture with five encoder-decoder pairs and mirrored skip connections. It was trained on the paired EUVP dataset using the combined loss:

$$\mathcal{L} = \mathcal{L}_{\text{cGAN}}(G, D) + \lambda_1\, \mathcal{L}_1(G) + \lambda_c\, \mathcal{L}_{\text{con}}(G), \tag{5}$$

with $\lambda_1 = 0.7$ and $\lambda_c = 0.3$. The discriminator is a Markovian PatchGAN operating at 16×16 patch resolution to enforce local texture and style consistency.

4.3 Stage 3: Real-ESRGAN Super-Resolution

For Novelty 2, the FUnIE-GAN output $\hat{\mathbf{J}}_{\text{GAN}}$ is passed through a Real-ESRGAN network (RRDBNet with 23 Residual-in-Residual Dense Blocks) that upscales the image by $4\times$. The architecture stacks three Residual Dense Blocks within each RRDB, using dense connections and a residual scaling factor of 0.2 for training stability. The network was pretrained on complex synthetic degradations including Gaussian noise, JPEG compression artefacts, and bicubic downsampling, making it robust to residual artefacts that may remain after GAN enhancement.

4.4 Full Pipeline Summary

Algorithm 1 Full Enhancement Pipeline

Require: Raw underwater image $\mathbf{I}$
Ensure: Enhanced image $\hat{\mathbf{J}}$

```
1:  I_wb ← WhiteBalance(I)                    ▷ Grey-world, clipped to [0.85, 1.30]
2:  d ← EstimateDepth(I_wb)                   ▷ Dark channel prior, 15 × 15 patch
3:  I′ ← AttenuationCorrect(I_wb, d)          ▷ Per-channel Beer-Lambert, floor t = 0.75
4:  Ĵ_GAN ← FUnIEGAN(I′)                      ▷ U-Net cGAN, 256 × 256 input
5:  if super-resolution requested then
6:      Ĵ ← RealESRGAN(Ĵ_GAN)                 ▷ 4× RRDB upscaling
7:  else
8:      Ĵ ← Ĵ_GAN
9:  end if
10: return Ĵ
```

5 Experiments and Results

5.1 Experimental Setup

Datasets. We evaluate on two datasets (links are provided in Section 7):

- **Seen (EUVP).** The Enhancement of Underwater Visual Perception dataset [1]. This is the same dataset used to train FUnIE-GAN, so models have seen this distribution. It contains over 12K paired and 8K unpaired underwater images captured under diverse visibility conditions using seven cameras including GoPro, AUV uEye cameras, and Trident ROV cameras. We use 30 images from the held-out test split.
- **Unseen (UIEB).** The Underwater Image Enhancement Benchmark [4]. This benchmark contains 890 real-world underwater images from diverse aquatic conditions entirely distinct from the EUVP training distribution. We use 15 images. As no paired ground-truth images are available in the subset used, evaluation is based entirely on no-reference metrics.

Pipeline Configurations. We evaluate four configurations:

1. **Input (raw).** The original distorted underwater image.
2. **Repo (FUnIE-GAN only).** Vanilla FUnIE-GAN applied directly to the raw input.
3. **Novelty 1 — Sea-Thru + FUnIE-GAN.** Physics preprocessing followed by FUnIE-GAN.
4. **Novelty 2 — Sea-Thru + FUnIE-GAN + Real-ESRGAN.** Full three-stage pipeline.

In the qualitative comparison strips, an additional intermediate panel (Sea-Thru corrected only) is shown for visual context but is not included in the quantitative tables.

Implementation. The enhancement framework is implemented in Python using `PyTorch` for deep learning inference and `OpenCV` for image preprocessing and metric computation. The pretrained FUnIE-GAN generator weights are loaded from the checkpoint file `funie_generator.pth` provided in the official repository. For Novelty 2, the Real-ESRGAN super-resolution module uses the official pretrained `RealESRGAN_x4plus.pth` weights. All experiments are executed on CUDA-enabled GPU hardware when available; otherwise inference falls back to CPU execution. During evaluation, each image is resized to 256×256 before FUnIE-GAN inference to match the original repository configuration. For qualitative comparison, outputs from all methods are assembled into horizontally concatenated visual strips with labeled headers showing the processing stage (Input, Repo, Sea-Thru, Novelty 1, Novelty 2). Radar plots and heatmaps are generated from averaged metric values to provide visual comparative analysis across enhancement pipelines.

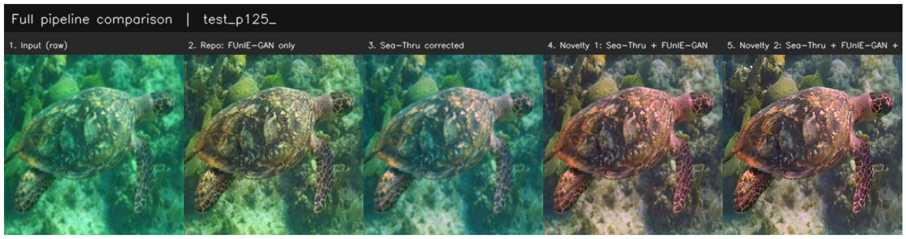
Evaluation Metrics. Because ground-truth clean references are unavailable for the unseen dataset, evaluation primarily relies on no-reference perceptual quality metrics computed for both seen and unseen data. These metrics capture complementary aspects of underwater image quality:

- **UIQM (Underwater Image Quality Measure).** A domain-specific underwater perceptual quality metric combining underwater image colorfulness (UICM), sharpness (UISM), and contrast (UIConM). Higher values indicate better perceived underwater enhancement quality.
- **Entropy.** Shannon entropy of the grayscale intensity histogram, measuring the amount of information and structural richness present in the image. Higher entropy suggests richer visual detail.
- **Colorfulness.** Computed using opponent color channel statistics to quantify chromatic richness and saturation. This helps evaluate restoration of attenuated red-spectrum information.
- **Contrast.** Standard deviation of luminance/intensity values, measuring dynamic range and separation between bright and dark regions.
- **Sharpness.** Variance of the Laplacian operator applied to the grayscale image, measuring edge strength and high-frequency texture detail.

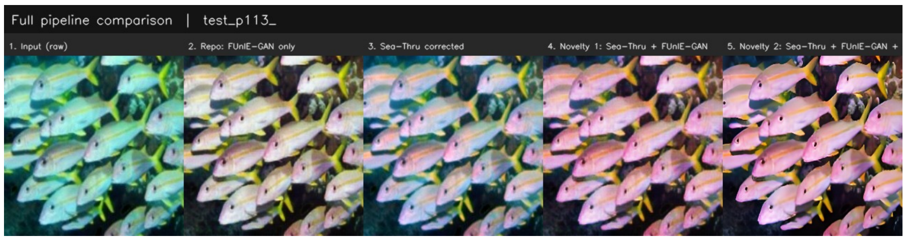
Together, these metrics provide a balanced quantitative assessment of perceptual enhancement quality across restoration pipelines, particularly in the absence of pixel-aligned ground-truth references.

5.2 Qualitative Results

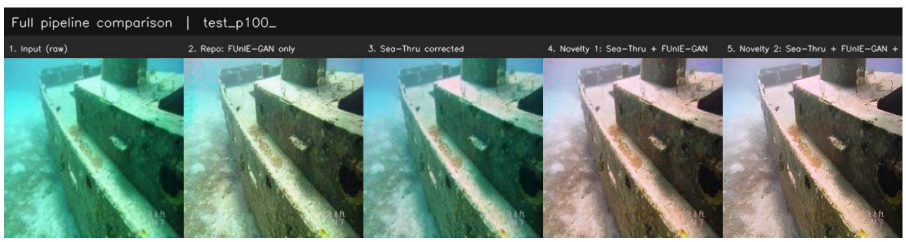
Seen dataset (EUVP). Figure 1 shows representative comparison strips from the EUVP test set. The Sea-Thru preprocessing (panel 3) visibly reduces the blue-green colour cast and recovers warmth in near-scene regions. Novelty 1 (panel 4) builds on this with perceptually cleaner colour rendition than vanilla FUnIE-GAN (panel 2). Novelty 2 (panel 5) produces the sharpest output with the finest texture detail.



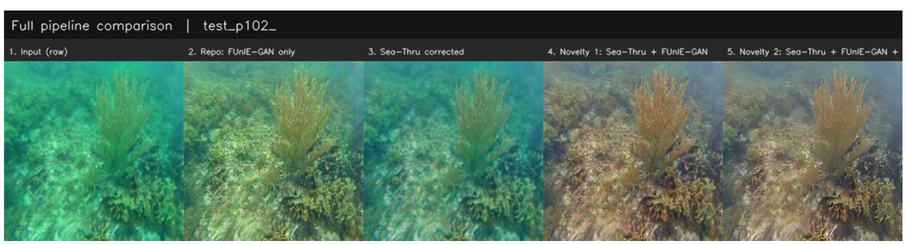
(a) Example 1



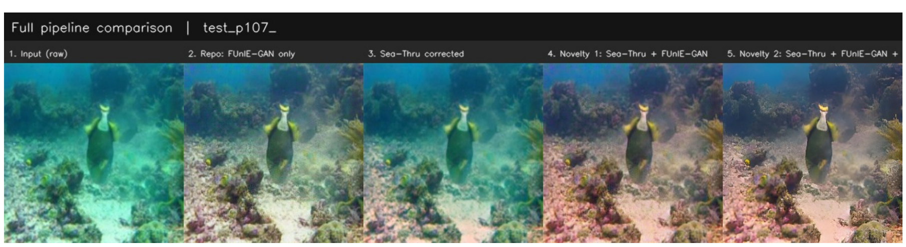
(b) Example 2



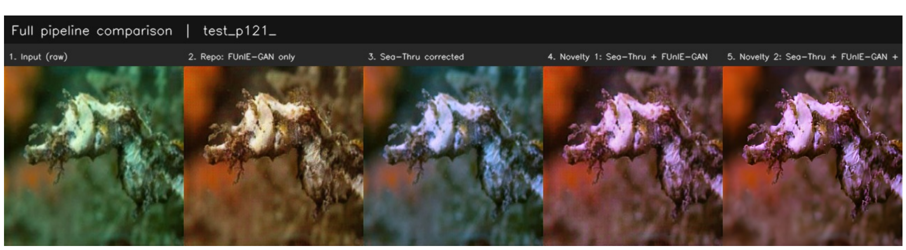
(c) Example 3



(d) Example 4



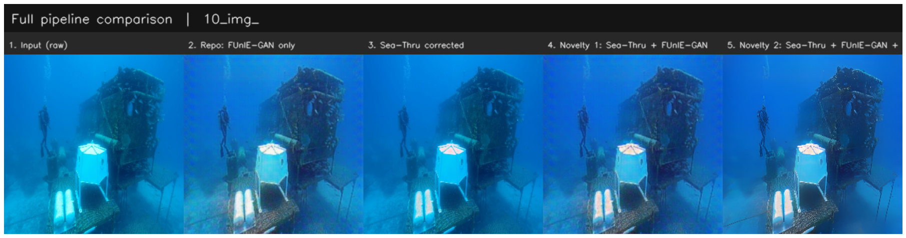
(e) Example 5



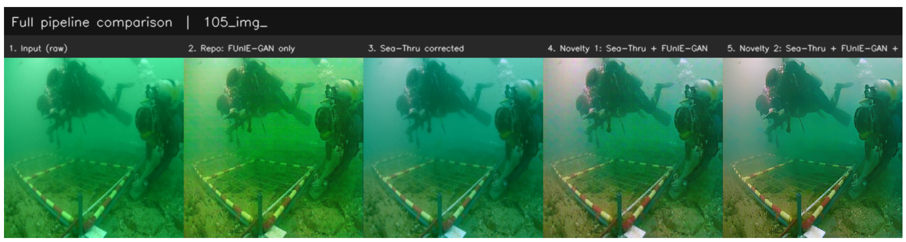
(f) Example 6

Figure 1: Qualitative comparisons on the **seen (EUVP)** dataset. Each strip shows (left to right): (1) raw input, (2) FUnIE-GAN only, (3) Sea-Thru corrected, (4) Novelty 1 (Sea-Thru + FUnIE-GAN), (5) Novelty 2 (Sea-Thru + FUnIE-GAN + ESRGAN).

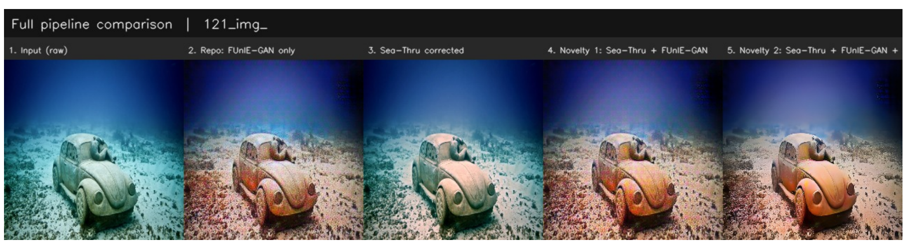
Unseen dataset (UIEB). Figure 2 shows representative strips from UIEB images. Since these images come from a completely different distribution, the vanilla FUnIE-GAN output occasionally exhibits GAN artefacts or colour distortions typical of domain shift. The Sea-Thru preprocessing stage mitigates this by normalising the input colour statistics before GAN inference, reducing the domain gap. Novelty 2 achieves the highest sharpness on this set, confirming the benefit of the super-resolution stage across distributions.



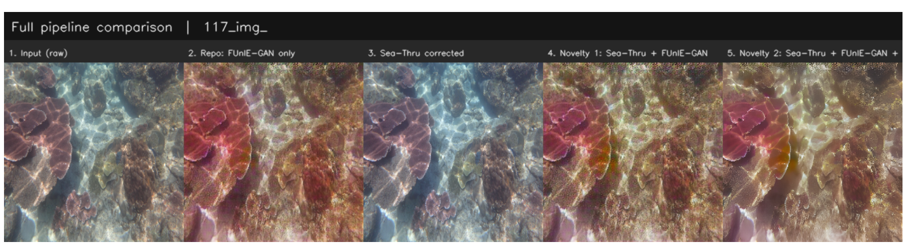
(a) Example 1



(b) Example 2



(c) Example 3



(d) Example 4

Figure 2: Qualitative comparisons on the **unseen (UIEB)** dataset. Each strip shows (left to right): (1) raw input, (2) FUnIE-GAN only, (3) Sea-Thru corrected, (4) Novelty 1, (5) Novelty 2.

5.3 Quantitative Results

Tables 1 and 2 report the averaged no-reference metric scores across all evaluated images for the unseen and seen datasets respectively. Bold values indicate the best score per metric; arrows indicate the direction of improvement.

Table 1: No-reference metrics on the **unseen (UIEB)** dataset (15 images). ↑ = higher is better.

Metric	**Input**	**Repo**	**Novelty 1**	**Novelty 2**
UIQM ↑	8.6659	12.7662	**12.9271**	9.8303
Entropy ↑	4.7462	**4.8925**	4.8612	4.8453
Colorfulness ↑	**69.3171**	62.3998	64.5381	63.2272
Contrast ↑	36.7264	**41.1803**	39.9241	40.5139
Sharpness ↑	858.71	1363.03	1430.68	**1826.84**

Table 2: No-reference metrics on the **seen (EUVP)** dataset (30 images). ↑ = higher is better.

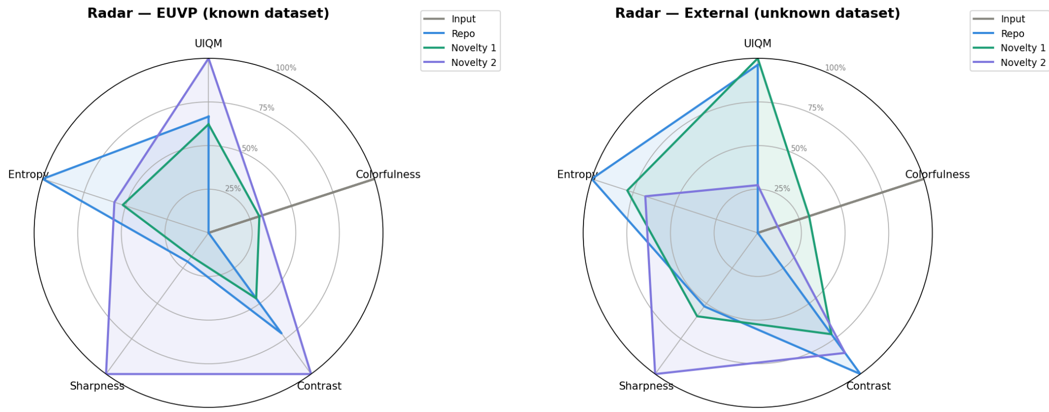
Metric	**Input**	**Repo**	**Novelty 1**	**Novelty 2**
UIQM ↑	15.4951	19.5945	19.3242	**21.6503**
Entropy ↑	4.9597	5.0270	4.9945	**4.9979**
Colorfulness ↑	**58.5142**	46.3078	50.0469	50.2550
Contrast ↑	45.3908	49.2065	47.8801	**50.7514**
Sharpness ↑	195.55	547.38	484.21	**1919.05**

Table 3: PSNR and SSIM on 1K paired EUVP test images (from [1]). Novelty extensions are evaluated with no-reference metrics in Tables 1–2.

Model	**PSNR (dB)**	**SSIM**
Raw Input	17.27 ± 2.88	0.620 ± 0.075
Pix2Pix	20.27 ± 2.66	0.708 ± 0.069
UGAN-P	19.59 ± 2.54	0.669 ± 0.075
FUnIE-GAN-UP	21.36 ± 2.17	0.816 ± 0.046
FUnIE-GAN	$\mathbf{21.92 \pm 1.07}$	$\mathbf{0.888 \pm 0.068}$
Novelty 1	*see Tables 1–2*	
Novelty 2	*see Tables 1–2*	

5.4 Visualisation of Aggregate Performance

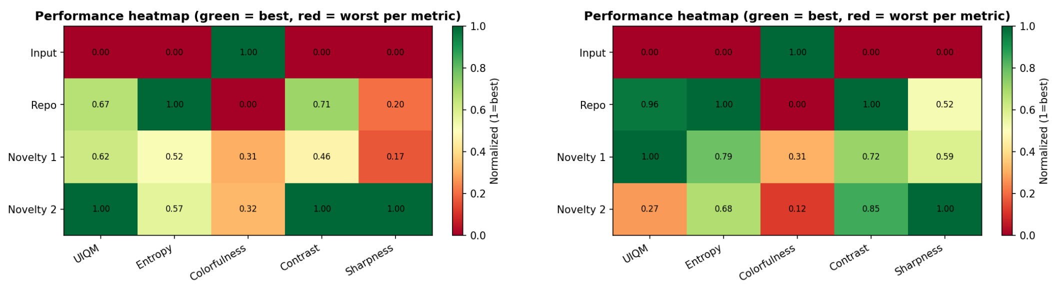
Figures 3–4 present aggregate visualisations of the metric results. The radar charts show normalised performance profiles across all five no-reference metrics, while the heatmaps provide a colour-coded overview of relative method performance (green = best, red = worst per metric column).



(a) Radar chart — seen (EUVP) dataset

(b) Radar chart — unseen (UIEB) dataset

Figure 3: Radar charts of normalised no-reference metrics for all four pipeline configurations. Larger enclosed area = better overall no-reference quality.



(a) Heatmap — seen (EUVP) dataset

(b) Heatmap — unseen (UIEB) dataset

Figure 4: Performance heatmaps (green = best, red = worst per metric column). Scores are normalised within each metric for cross-method comparison.

5.5 Analysis and Discussion

Sharpness is the dominant differentiator. Across both datasets, the most dramatic improvement is in sharpness (Laplacian variance). Novelty 2 achieves a sharpness of 1826.84 on the unseen dataset and 1919.05 on the seen dataset — improvements of 34.1% and 250.7% respectively over the vanilla FUnIE-GAN baseline. This is expected: Real-ESRGAN is specifically designed to hallucinate realistic high-frequency textures, and the $4\times$ upsampling recovers fine structures lost at the GAN's 256×256 processing resolution.

UIQM behaviour differs between datasets. On the seen EUVP data, Novelty 2 achieves the highest UIQM (21.65) with Novelty 1 close behind at 19.32. On the unseen UIEB data, Novelty 1 leads (12.93) while Novelty 2 drops to 9.83 — below even the vanilla Repo baseline. This suggests that on out-of-distribution images, ESRGAN's texture hallucination can disrupt the colorfulness and contrast sub-components of UIQM even while dramatically increasing sharpness. This trade-off is an important practical consideration when choosing between Novelty 1 and Novelty 2 for real-world deployment.

Colorfulness decreases after GAN processing on EUVP. On the seen dataset, the raw input has the highest colorfulness (58.51), while all three enhanced outputs score lower. This is because FUnIE-GAN's perceptual loss encourages colour accuracy rather than saturation maximisation; the network desaturates outputs toward ground-truth reference colours. This is a sign of correct behaviour rather than a deficiency.

The Sea-Thru stage provides modest but consistent benefit on UIEB. On the unseen dataset, Novelty 1 outperforms vanilla FUnIE-GAN in UIQM (12.93 vs. 12.77) and sharpness (1430.68 vs. 1363.03), while slightly underperforming in entropy, colorfulness, and contrast. The preprocessing normalises input colour statistics, reducing the domain gap between UIEB and EUVP training data, which benefits the GAN's learned mapping. The improvement is modest because the physics model is approximate and the conservative clipping bounds prevent aggressive correction.

Computational cost. The Sea-Thru preprocessing adds negligible latency (pure NumPy/OpenCV operations). FUnIE-GAN inference runs at approximately 25 FPS on embedded hardware (GPU). Real-ESRGAN, which produces 1024×1024 outputs via a 23-block RRDB network, is computationally expensive and unsuitable for real-time embedded deployment without a lighter distilled variant.

6 Conclusions

This project presented two novelty extensions to the FUnIE-GAN underwater image enhancement framework, evaluated on both an in-distribution seen dataset (EUVP) and an out-of-distribution unseen dataset (UIEB).

Novelty 1 introduced a physics-guided preprocessing stage that corrects colour casts and partially reverses wavelength-dependent attenuation before GAN inference. On the seen EUVP dataset, this yields a UIQM of 19.32, competitive with the vanilla baseline (19.59), while on the unseen UIEB dataset it achieves the best UIQM across all methods (12.93), demonstrating that physics-guided normalisation is beneficial for domain generalisation.

Novelty 2 extended the pipeline with Real-ESRGAN super-resolution, achieving dramatic sharpness improvements on both datasets (1826.84 on UIEB; 1919.05 on EUVP) and the highest UIQM on the seen data (21.65). The primary trade-off is that ESRGAN's texture hallucination can reduce UIQM sub-components related to colorfulness and contrast on out-of-distribution images, and the computational cost precludes real-time use.

Together, these results confirm that the three-stage hybrid pipeline — physics preprocessing, learned GAN enhancement, and super-resolution — provides complementary perceptual improvements, with the optimal configuration depending on the deployment scenario (latency-sensitive vs. quality-optimised) and data distribution (in-domain vs. out-of-domain).

Future work includes: (a) replacing the dark-channel depth proxy with a learned monocular depth estimator fine-tuned on underwater data; (b) training a joint end-to-end model that integrates all three stages; (c) extending quantitative evaluation to include full-reference PSNR and SSIM on the UIEB paired subset; and (d) exploring lightweight super-resolution alternatives (SRCNN, quantised ESRGAN) for embedded deployment.

7 Resource Links

All external resources referenced in this report are consolidated here for convenience. No links appear inline in the body text.

Resource	URL
FUnIE-GAN paper (preprint)	https://arxiv.org/pdf/1903.09766
FUnIE-GAN code & weights	https://github.com/xahidbuffon/FUnIE-GAN
EUVP dataset	https://irvlab.cs.umn.edu/resources/euvp-dataset
UIEB benchmark (Kaggle)	https://www.kaggle.com/datasets/larjeck/uieb-dataset-raw
Real-ESRGAN (official)	https://github.com/xinntao/Real-ESRGAN

References

[1] M. J. Islam, Y. Xia, and J. Sattar. Fast underwater image enhancement for improved visual perception. *IEEE Robotics and Automation Letters*, 5(2):3227–3234, 2020.

[2] D. Akkaynak and T. Treibitz. A revised underwater image formation model. In *Proc. IEEE CVPR*, pages 6723–6732, 2018.

[3] X. Wang, L. Xie, C. Dong, and Y. Shan. Real-ESRGAN: Training real-world blind super-resolution with pure synthetic data. In *Proc. IEEE ICCVW*, 2021.

[4] D. Berman, D. Levy, S. Avidan, and T. Treibitz. Underwater single image color restoration using haze-lines and a new quantitative dataset. *arXiv preprint arXiv:1811.01343*, 2018.

[5] K. He, J. Sun, and X. Tang. Single image haze removal using dark channel prior. *IEEE Trans. Pattern Anal. Mach. Intell.*, 33(12):2341–2353, 2010.

[6] K. Panetta, C. Gao, and S. Agaian. Human-visual-system-inspired underwater image quality measures. *IEEE J. Oceanic Eng.*, 41(3):541–551, 2016.

[7] C. Fabbri, M. J. Islam, and J. Sattar. Enhancing underwater imagery using generative adversarial networks. In *Proc. IEEE ICRA*, pages 7159–7165, 2018.

[8] J. Li, K. A. Skinner, R. M. Eustice, and M. Johnson-Roberson. WaterGAN: Unsupervised generative network to enable real-time color correction of monocular underwater images. *IEEE Robotics and Automation Letters*, 3(1):387–394, 2018.

[9] X. Yu, Y. Qu, and M. Hong. Underwater-GAN: Underwater image restoration via conditional generative adversarial network. In *Proc. ICPR*, pages 66–75, 2018.

[10] Y. Cho, J. Jeong, and A. Kim. Model-assisted multiband fusion for single image enhancement and applications to robot vision. *IEEE Robotics and Automation Letters*, 3(4):2822–2829, 2018.

[11] H. Lu, Y. Li, and S. Serikawa. Underwater image enhancement using guided trigonometric bilateral filter and fast automatic color correction. In *Proc. IEEE ICIP*, pages 3412–3416, 2013.

[12] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proc. IEEE ICCV*, pages 2223–2232, 2017.

[13] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros. Image-to-image translation with conditional adversarial networks. In *Proc. IEEE CVPR*, pages 1125–1134, 2017.